

Subject Description Form

Subject Code	AAE5109
Subject Title	Advanced Neuro-ergonomics and Cognitive Science in Aviation
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. Systematically understand the core theories and methodologies of neuro-ergonomics and cognitive science in aviation. 2. Explore how to integrate cutting-edge neuro-ergonomics technologies, human factors experiments, and cognitive analysis into aviation safety. 3. Cultivate problem-solving ability grounded in cognitive science to bridge the gap from data analysis to real-world interventions in aviation safety situations.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Identify the aviation personnel's cognitive problems and correlate them with task operational interactions in aviation. b. Highlight the complex problems related to cognition and design the neuro-ergonomics-involved solutions/experiments to address aviation-specific cognitive challenges. c. Apply statistical tools/AI algorithms to do neuro-psycho-physiological data analysis for cognitive patterns investigation and cognitive modelling. d. Be aware of what/how to realize the cognitive science and propose tailored cognitive applications in aviation.
Subject Synopsis/ Indicative Syllabus	Introduction: The overall thought, principles, and applications of neuro-ergonomics and cognitive science in aviation. Human Cognition in Aviation: Cognitive workload, situational awareness, fatigue, decision-making, attention, and stress of air traffic controllers and pilots. Data Analysis: Data pre-processing, cleaning, interpreting, and visualizing, and pattern recognition of neuro-psycho-physiological data. Technological Advancements in Neuro-Ergonomics: Cognitive states modelling, monitoring, prediction, neuroimaging, neurofeedback, and brain-computer interfaces. Laboratory and Case Studies: Neuro-ergonomic technologies and simulators tutorial. Human cognitive workload monitoring and situation awareness detection.

Teaching/Learning Methodology	Teaching is conducted through class lectures, laboratory, and case studies. The basic knowledge, research methodology and theoretical models will be introduced. The understanding of how to address and identify the human factors problem and formulate the resolution will be emphasized. Research methodology, case study and analytics skills and tools (e.g., MATLAB) are taught in class as well as the related real-life scenarios to enhance the teaching and learning abilities.																																													
	Teaching/Learning Methodology		Outcomes																																											
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	Lecture		√	√	√	√																																								
	Laboratory			√	√																																									
	Case Study			√	√	√																																								
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>1. Assignment</td><td>15%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>2. Laboratory</td><td>15%</td><td></td><td>√</td><td>√</td><td></td></tr><tr><td>3. Mini project</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Final examination</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.6 × Continuous Assessment + 0.4 × Final Examination The continuous assessment (60%) is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus via assignment, laboratory and mini project. The assignment will test students' understanding of the basic concepts of data mining, neuro-ergonomics, and cognitive science in aviation. The laboratory will enhance students' practical skills in using neuro equipment and test their ability to collect accurate neuro data with neuro equipment. The mini project will assess students' teamwork, self-learning, problem identification, and presentation skills in implementing neuro-ergonomic solutions to enhance aviation safety and efficiency. The final examination (40%) will also be considered to assess the students’ learning outcome.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Assignment	15%	√	√			2. Laboratory	15%		√	√		3. Mini project	30%	√	√	√	√	4. Final examination	40%	√	√	√	√	Total	100%				
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Student Study Effort Expected	Class contact:	
	▪ Lecture/Case Study	33 Hrs.
	▪ Laboratory	6 Hrs
	Other student study effort:	
	▪ Self-learning/preparation	36 Hrs.
	▪ Literature study/case study/reading	36 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	1. van Weelden, E., Alimardani, M., Wiltshire, T. J., & Louwerse, M. M. (2022). Aviation and neurophysiology: A systematic review. <i>Applied ergonomics</i>, 105, 103838. 2. Aricò, P., et al., Human factors and neurophysiological metrics in air traffic control: a critical review. <i>IEEE Reviews in Biomedical Engineering</i>, 2017. 10: p. 250-263. 3. Cohen, M.X., <i>Analyzing neural time series data: theory and practice</i>. 2014: MIT press. 4. Parasuraman, R., Lee, J. D., & Kirlik, A. (2013). Neuroergonomics: brain-inspired cognitive engineering. In <i>The Oxford Handbook of Cognitive Engineering</i> (pp. 159-177). Oxford: Oxford University Press. 5. Klaproth, O.W., C. Vernaleken, L.R. Krol, M. Halbruegge, T.O. Zander, and N. Russwinkel, Tracing Pilots' Situation Assessment by Neuroadaptive Cognitive Modeling. <i>Frontiers in neuroscience</i>, 2020. 14: p. 795-795. 6. Adams, F., Pessoa Jr, O., & Kogler Jr, J. E. (Eds.). (2017). <i>Cognitive science: Recent advances and recurring problems</i>.	

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